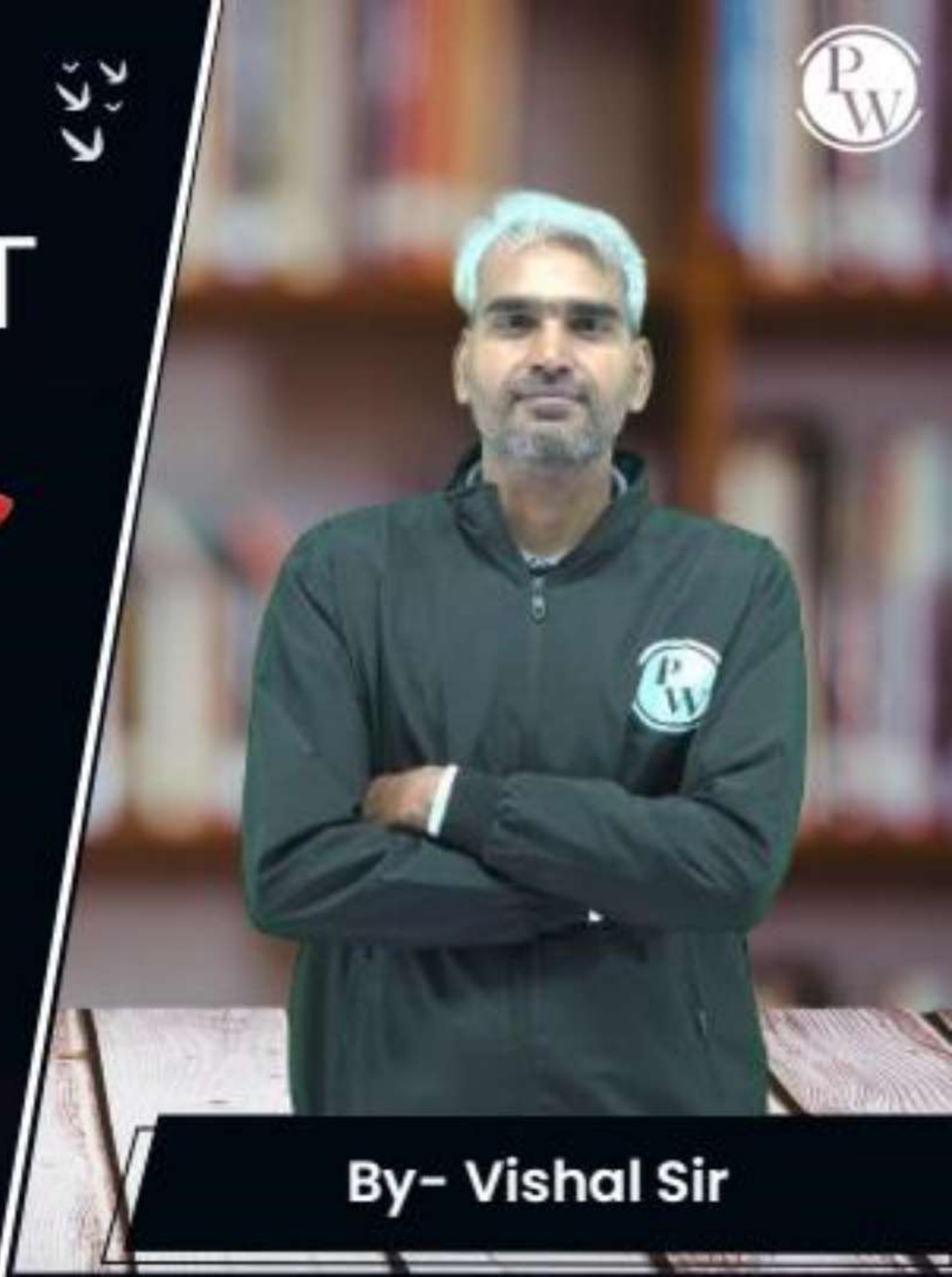
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 08



By- Vishal Sir

Recap of Previous Lecture



- Topic: Minimal Cover (Canonical Cover)
- Topic: Procedure to obtain minimal cover for a given FD set

Slide

Topics to be Covered



- **Topic** Schema refinement (Normalization)
- **Topic** Problems because of redundancy in a relation
- **Topic** Properties of decomposition



Topic : Schema refinement (Normalization)

Normalization is a process of reducing / eliminating the redundancy present in the relational table



Topic : Redundancy in a relation

* If independent informations are stored in a single relational table, then redundancy is possible.

Topic : Redundancy in a relation

Student information

Course information

Enroll

Sid	Sname	Cid	Cname
S_1	A	C_1	OS
S_1	A	C_2	DBMS
S_3	A	C_2	DBMS
S_4	B	C_2	DBMS
S_4	B	C_3	OS

Sid → Sname

Sname ↛ Sid

Cid → Cname

Cname ↛ Cid

Redundancy {duplication of known information}

Two FDs holds true in this relation
Sid → Sname
& Cid → Cname
∴ C.K. = (Sid, Cid)

* Sid & Cid are the attributes of Primary key
∴ Their value can not be NULL

It is the only candidate key of the Relⁿ
∴ It will also be the Primary key

it is not redundancy, it is useful information
It tells us that name of Student with Sid = S_1 is A, as well as name of student with Sid = S_2 is also 'A'.

* If redundancy is present in the relation, then it may cause various problems



Topic : Problems because of redundancy

→ (1) It requires more storage space in the disk.

Serious Problems
- (2) Insertion Anomaly
- (3) Deletion Anomaly
- (4) Updation Anomaly

Insertion anomaly :- If independent informations are stored in a single table then sometimes it may not be possible to insert one information into the relational table without inserting other independent informations.

eg: In the enroll table we can not insert the information of a new course (C4, DSAT) until some students enroll for that course, because we can not set the value of Sid to NULL in any tuple as "Sid" is a part of Primary key.

Deletion Anomaly :- If we try to delete one independent information then we may loose some other independent informations from the table.

eg: If we try to delete the information of Student S_1, then we can not simply delete the Student information without deletion of Course information associated with that Student, because Sid field can not be NULL (Sid is part of Primary Key), We will have to delete the complete tuples associated with Sid = S_1, if we do so then we will loose the information of Course C_1 {information of Course C_2 will still be available in the table because other Students have enrolled for Course C_2.}

→ Updation Anomaly: Updation will be required in all duplicate copies and it will be time consuming.

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Topic : Schema refinement (Normalization)

with redundancy

Normalization is a process of decomposing (splitting) a relational tables into smaller tables (sub-relations) such that it eliminates/reduces the data redundancy, and it can overcome undesirable characteristics like Insertion, Updation and Deletion Anomalies

Topic : Schema refinement (Normalization)

After this decomposition,

① Redundancy is eliminated

and

② All the anomalies are overcome, because there is no redundancy

Sid	Sname	Cid	Cname
S_1	A	C_1	OS
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	OS

Sid → Sname
Cid → Cname

let, this relation is decomposed into three sub-relations

Student

Sid	Sname
S_1	A
S_2	A
S_3	B

it is not redundancy

Sid → Sname

Course

Cid	Cname
C_1	OS
C_2	DBMS
C_3	OS

Not Redundancy

Cid → Cname

Enroll

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_2
S_3	C_2
S_3	C_3

it is Not Redundancy



Topic : Properties of decomposition

While decomposing a relational table into sub-relations (i.e., smaller tables) then, we must ensure that following two properties are satisfied

1. Decomposition must be dependency preserving decomposition
   { All the dependencies present in the original relation must also be available even after the decomposition of relation }
2. Decomposition must be lossless Join decomposition
   { It states that, if we join (⋈) the sub-relations obtained after the decomposition then, we must get the exact same data of original relation. }

Ultimately, there should not be any type of information loss.
FD and/or data



2 mins Summary

Topic Schema refinement (Normalization)

Topic Problems because of redundancy in a relation

Topic Properties of decomposition

THANK - YOU